

## UNC304: DISCRETE MATHEMATICS

| L | T | P | Cr  |
|---|---|---|-----|
| 3 | 0 | 0 | 3.0 |

**Course Objectives:** To understand mathematical reasoning in order to read, comprehend and construct Mathematical arguments as well as to solve problems, occurred in the development of programming languages. To work with discrete structures such as graphs to study the structure of the world wide web, to model a computer network and to find the shortest path between two places in a transportation network.

**Basic Logic:** Propositional logic, Logical connectives, Truth tables, Disjunctive normal form, Validity of a well-formed formula, Propositional inference rules (e.g., modus ponens, modus tollens), Universal and existential quantifiers and their negations.

**Sets, Relations and Functions:** Methods for describing a set, e.g., listing elements, set builder notation, Venn diagrams, Union, intersection, set difference, complement, Cartesian product, Power sets, Cardinality of finite sets, Reflexivity, symmetry, antisymmetry, transitivity, equivalence relations, partial orders, Poset, Lattice, Hasse's Diagram, Domain, target, and range/image of a function, Surjections, injections, bijections, Inverses, Composition.

**Proof Techniques:** Notions of implication, double implication, equivalence, converse, inverse, contrapositive, negation, and contradiction, The structure of mathematical proofs: Direct proofs, Disproving by counterexample, Proof by contrapositive, Proof by contradiction, Induction over natural numbers, Weak and strong induction (i.e., First and Second Principle of Induction), Recursive mathematical definitions, Well orderings.

**Basics of Counting:** Counting arguments, Set cardinality and counting, Sum and product rules, Inclusion-exclusion principle, Arithmetic and geometric progressions, The pigeonhole principle, Permutations and combinations, Basic definitions, The binomial theorem, Recurrence relations.

**Graph Theory:** Properties, Traversal strategies, Undirected graphs, Directed graphs, Weighted graphs, Spanning trees/forests, Graph isomorphism, Eulerian cycle and Hamiltonian cycle, adjacency and incidence matrices, vertex colouring, planarity, trees.

**Familiarization with standard:** Formal Methods and Models for System Design: IEEE 1596-1992, Graphical Symbols for Logic Functions: IEEE 91-1984, Discrete Mathematics and Algorithmic Reasoning Support Standards (conceptual alignment): IEEE 1012-2016.

### Course Learning Outcomes:

1. Apply Knowledge of mathematics in truth tables and the rules of propositional & predicate calculus
2. Apply the logical and analytical structure of proofs and work symbolically with connectives and quantifiers to produce logically valid, correct and clear arguments.
3. Evaluate elementary mathematical arguments and identify fallacious reasoning
4. Analyze real-world problems using graph theory.

### Text Books

1. Discrete Mathematics and its Applications by Kenneth H Rosen

### Reference Books

1. Discrete and Combinatorial Mathematics by Ralph P Grimaldi
2. Elements of Discrete Mathematics, C. L. Liu

## UECxxx: ELECTRONICS DEVICES AND CIRCUITS

| L | T | P | Cr  |
|---|---|---|-----|
| 3 | 1 | 2 | 4.5 |

### Course Objective:

This course introduces the fundamental principles of semiconductor devices, beginning with PN junctions and BJT operation, then focusing extensively on MOSFET device operation, CMOS digital design, and CMOS fabrication processes.

### Syllabus

**Introduction To Semiconductor:** Energy bands in semiconductors, types of semiconductors, charge carriers, intrinsic, extrinsic semiconductor, mass action law, p-n junction diode, energy band diagram, Zener diode. BJT operation, types and its biasing configurations, BJT Switch, FET.

**NMOS & CMOS Process Technology:** MOS fabrication process steps: Masking sequence of NMOS and CMOS structures, latch up in CMOS, electrical design rules, stick diagram and layout.

**MOS Transistor Theory:** MOS structure, types and its operation, I-V characteristics, threshold voltage equation, body bias effect, second order effects, scaling theory, short-channel effects, MOS Capacitors, MOS switch, Advanced MOS Structures: FinFET, GAA FETs, Nanosheet Transistors – Concepts and Comparisons.

**Circuit Characterization:** Resistive load inverter and CMOS inverter: static characteristics, switching characteristics, static and dynamic power dissipation, noise margin.

**Combinational Circuits:** CMOS logic styles, realization of simple gates, complex logic circuits, pass gate and transmission gate.

**Familiarization with standard:** IEEE 1181-1991.IEEE 1801-2024.

**Laboratory Work:** Familiarization with circuit design/simulation tools (Cadence/Mentor/Tanner Tools) for schematic and layout entry, circuit simulation using SPICE. DC transfer characteristics of inverters, transient response, calculating propagation delays, rise and fall times, circuit design of inverters, complex gates with given constraints.

**Course learning outcome (CLO):** The student will be able to:

1. Understand the working of PN junction diode and MOS transistor operation.
2. Analyze the static and dynamic characteristics of Resistive load Inverter and CMOS inverter.
3. Implement the various CMOS logic circuits.
4. Illustrate the fabrication process for NMOS and CMOS circuits.

### Text Books:

1. Kang, Sung-Mo (Steve) & Leblebici, Yusuf., CMOS Digital Integrated Circuits Analysis & Design, McGraw Hill, (1999) 2nd ed.
2. A. S. Sedra and K. C. Smith, Microelectronic Circuits. 4th ed. New York, NY: Oxford University Press, 1998.

### Reference Books:

1. Gregorian, R. and Temes, G.C., Analog MOS Integrated Circuits for Signal Processing, John Wiley (2004).
2. Jan Rabaey, A. Chandrakasan & Nikolic, B., Digital Integrated Circuits – A Design Perspective, Pearson, (2003) 2nd ed.
3. CMOS VLSI Design: A Circuits and Systems Perspective, 4th ed., Neil Weste and David Harris, Pearson Addison Wesley, 2011.
4. Pucknell D. A., & Eshraghian, K., Basic VLSI Design, Prentice Hall of India, (2007) 3rd ed